

Linear systems and elimination

Monotonic optimal backward error along LSMR

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Independently reviewed resolution - 2026-09-11

Negative resolution. Sections 1-4 give one exact full-column-rank 4×3 LSMR example for which both displayed errors increase from the first to the second nonzero iterate. The matrix-only spectral backward error satisfies $\mu(x_1)^2 \leq 1979/2000 < 99/100 < \mu(x_2)^2$, and the specified approximation also strictly increases. The right-hand side stays fixed. This settles the canonical spectral-norm formulation; the cited SISC paper uses a different default norm convention, so no Frobenius-error conclusion is inferred.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#), [independent proof review](#), and [submission record](#). The supplied notes were reconstructed with substantial AI assistance. This is independent agent verification, not external human peer review or formal proof-assistant certification; no novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects monotonicity of an optimization-defined error along coupled Krylov iterates; community impact is a stopping and reliability guarantee for a widely used least-squares method.

Lean proof and verification evidence — 2026-09-14

Both original monotonicity claims have Lean-verified negative answers. The complete pair of monotonicity claims is disproved by the exact first two nonzero iterates of undamped, zero-initial-guess LSMR on a real 4-by-3 system. The formalization proves the genuine minimum-length normal-residual minimizers over the Krylov spaces, the matrix-only spectral backward-error infimum and attainment, and the prescribed Moore–Penrose projector approximation. Both errors strictly increase; the right-hand side remains fixed.

Immutable formal proof. The checked exports are `NLA.IE17.iterates`, `NLA.IE17.backward_increase`, `NLA.IE17.approximation_increase`, `NLA.IE17.counterexample`. Lean 4.33.1 uses pinned Mathlib `0df444a3` and LeanCert `621a43d7`; [complete pins and reproduction instructions](#) are retained. All four exported transitive axiom closures contain only `propext`, `Classical.choice` and `Quot.sound`.

Formalization: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final proof reviews passed. Kernel-only LeanCert and [fresh isolated Linux Comparator and kernel verification](#) passed, including rejection controls. [Original artifact](#), [exact checked inputs](#) and [independent operational audit](#) are retained. This is AI-assisted formal verification, not external human peer review or source-author endorsement.

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. In exact arithmetic, start LSMR at $x_0 = 0$: equivalently, x_k minimizes $\|A^T(b - Ax)\|_2$ over $\mathcal{K}_k(A^T A, A^T b)$. Work up to its exact termination and use its minimum-length iterate if necessary. Let $r = b - Ax$ and define the matrix-only normwise backward error

$$\mu(x) = \min\{\|E\|_2 : (A + E)^T((A + E)x - b) = 0\}.$$

For $x \neq 0$, put $K_x = [A^T, (\|r\|_2/\|x\|_2)I]^T$, $v_x = [r^T, 0^T]^T$, and

$$\tilde{\mu}(x) = \frac{\|K_x K_x^\dagger v_x\|_2}{\|x\|_2}.$$

Here $\dagger$ is the Moore–Penrose inverse; at an exact least-squares solution set both errors to zero.

Are both sequences $\mu(x_k)$ and $\tilde{\mu}(x_k)$ nonincreasing, for successive nonzero LSMR iterates? These two closely related claims are counted together, as in the source. The right-hand side b is kept fixed in the backward-error model.

A positive answer would justify backward-error stopping decisions without a later iteration making the current iterate less backward accurate. The known monotonicity of $\|r_k\|_2$ and $\|A^T r_k\|_2$ does not establish this statement.

References

D. C.-L. Fong, *Minimum-Residual Methods for Sparse Least-Squares Using Golub–Kahan Bidiagonalization*, Stanford dissertation 2011, §7.2.1 and definitions (4.1), (4.5), printed pp. 56–57, 118 ([primary PDF](#)). Fong and M. Saunders, *LSMR: An iterative algorithm for sparse least-squares problems*, SISC 33(2011), 2950–2971, §§6.1–6.2, Fig7.5 ([author PDF](#)). E. Hallman and M. Gu, *LSMB: Minimizing the backward error for least-squares problems*, SIMAX 39(2018), 1295–1317 ([author PDF](#)).

Status check — 2026-09-08

Exact-title and “LSMR optimal backward error monotonicity conjecture/proof/counterexample” searches found no resolution. The LSMB2018 discussion of nonmonotonicity concerns LSQR, and its proposed method differs from LSMR. Hallman’s May 2026 arXiv:2605.09211 develops backward-error formulas and estimates, without resolving this iterate-monotonicity conjecture. No recent explicit reaffirmation was located.

Audit update — 2026-09-10

Rechecked Fong’s [dissertation](#), §7.2.1, printed p. 118, which separately conjectures monotonicity of both displayed errors. [Hallman \(2026\)](#) studies a variational equation and lower bound for the backward error; it does not assert this iterate monotonicity. Searches for LSMR optimal-backward-error monotonicity located no proof or counterexample.